FINAL PROJECT REPORT

Sofia Lascano
BME 175
Partners: Calliope Foo, Jonathan Chung
Mentors: Brooke Longo, Edward Gordon
Instructor: Ying Chen

OBJECTIVE

The objective of this project was to design and manufacture a 3D-printed bioreactor to act as an *in vitro* model of the multi-segmental gastrointestinal tract, such that physiological conditions, microenvironments, and flow dynamics can be controlled and realistically simulated. This was done via computer-aided design, and the finalized printed product was tested for structural and mechanical integrity.

INTRODUCTION

The gastrointestinal tract represents one of the most complex and dynamic organ systems in the human body, comprising multiple interconnected segments that each perform specialized digestive, absorptive, and barrier functions. The human digestive system comprises the mouth, pharynx, oesophagus, stomach and small and large intestines, with the small intestine further divided into the duodenum, jejunum and ileum, each primarily involved in breaking down food, absorbing nutrients and water, and propelling food through the tract (Singh, 2024). Each segment of the GIT maintains unique physiological conditions that are individually regulated to optimize digestive processes and nutrient absorption.

The complexity of the GIT arises not only from its anatomical organization, but also from the distinct microenvironments that make up each region. Each component of the gastrointestinal tract exhibits unique characteristics that shape the structure and function of its microbiota. These segment-specific conditions include variations in pH levels ranging from highly acidic in the

stomach (pH 1.5-3.5) to neutral or slightly alkaline in the small intestine (pH 6-7.4), differences in enzyme concentrations, diverse microbial populations, and varied nutrient availability (García Mendez et al., 2023). Additionally, under physiological conditions the intestinal epithelium is exposed to one of the highest oxygen gradients in the body, a feature that is critical for maintaining proper epithelial function and regulating transport processes (Ward et al, 2014). All in all, it means that the cells all along the GIT system require precise control of their culture environment, including appropriate gas exchange, pH maintenance, and nutrient delivery to maintain their differentiated phenotypes and physiological functions.

In the field of medicine, studying the GIT is as important as it is challenging. *In vivo* methods are shrouded by ethical concerns if involving humans and/or live animals, and often very costly (Singh, 2024). Thus, the shift is becoming more marked towards *in vitro* models. *In vitro* intestinal models can potentially enable improved studies of cellular growth and proliferation, drug absorption, and host-microbial interactions, while reducing the aforementioned expense and ethical issues associated with the use of animal experiments (Costa et al., 2019). Traditional cell culture models, including two-dimensional monolayer cultures, fail to capture the three-dimensional architecture, dynamic flow conditions, and segment-specific microenvironments of the native GIT. Currently existing models that attempt to be more complex do not incorporate the entire gastrointestinal tract, with most models focusing on one or two components. Thus, we are presented with a challenge in accurately and precisely modeling one of the most intricate systems in the human body if we want to gain a better understanding of it to develop new solutions and combat disease.

Advanced bioreactor systems have emerged as a promising solution to overcome these limitations. Bioreactors are designed with the intention to control the different cell-culture

parameters, and to understand the effect of certain physical forces on cell and tissue behavior (Costa et al., 2019). These systems can provide controlled perfusion, maintain sterile conditions, monitor key parameters such as dissolved oxygen and pH, and simulate the mechanical forces present *in vivo*. This project thus aims to design such a bioreactor, with underlying simplifications, that could successfully model the different compartments of the GIT.

The bioreactor system designed in this project would address several needs in GIT modeling. First, its modular design allows for the connection of multiple intestinal compartments in series, enabling the simulation of sequential digestive processes as they occur *in vivo*. This modularity provides flexibility to add or remove compartments as needed for specific experimental designs, making the system adaptable to various research questions. Each compartment's conditions could then be tailored to match specific gaseous exchange levels, or be monitored for pH or other distinctive factors of the different GIT segments.

Second, the dual-flow capability of our bioreactor enables bidirectional media perfusion, which is useful for simulating the fluid dynamics of the GIT. This allows for the flow rates of media to mimic peristaltic movements and ensures adequate delivery of nutrients and oxygen while removing metabolic waste products. The ability to independently control flow parameters in each chamber enables the creation of segment-specific conditions that more closely replicate the physiological environment of different intestinal regions.

Third, the incorporation of three-dimensional silk scaffolds within each chamber provides a substrate for cell attachment and growth that better mimics the native extracellular matrix compared to flat culture surfaces. Silk scaffolds offer great biocompatibility, appropriate mechanical properties, and a porous architecture that supports cell infiltration and tissue-like organization (Chen et al., 2022). Emulating the natural topography of the small intestine with

biocompatible scaffolds can lead to improved cell growth and differentiation, as cells in a three-dimensional environment can experience nutrient and oxygen gradients that cells grown on flat surfaces cannot, which can affect their physiology including differentiation (Chen, 2015).

Finally, the transparency of the bioreactor material is a good feature to allow for real-time visual monitoring of cell growth, scaffold integrity, and fluid flow without disrupting the culture environment, which is also present for traditional Petri dish 2D cell culture. This non-invasive observation capability is important for assessing experimental progress and identifying potential issues during long-term perfusion studies.

By addressing the limitations of existing in vitro models through a modular, perfusion-capable, and transparent design, this bioreactor system provides a more physiologically relevant platform for studying the interconnected nature of the GIT. This work is part of a larger movement toward developing comprehensive *in vitro* models that can support advanced studies of digestion, nutrient absorption, drug metabolism, host-microbiome interactions, and gastrointestinal disease mechanisms in a controlled and reproducible manner.

METHODS

Bioreactor Materials Research

The selection of appropriate materials for bioreactor fabrication required balancing multiple performance criteria. Key requirements we considered included: (1) biocompatibility and non-cytotoxicity for cell culture applications, (2) sterilizability through autoclaving at 121-134°C without deformation, (3) sufficient mechanical strength to withstand tapping, drilling, clamping forces, and sustained internal pressure during long-term perfusion, (4) optical transparency for visual monitoring of scaffolds and flow, (5) low cell adhesion on chamber walls

to direct cell growth toward the silk scaffolds rather than bioreactor surfaces, and (6) resin availability for our specific printer model.

Three candidate resins were evaluated for use with the Formlabs Form 3B+ stereolithography printer available to us in the Tufts Science and Technology Center (SciTech). Clear Resin V4 offered high tensile strength (65 MPa) and cost-effectiveness, making it suitable for initial prototyping iterations. However, its lack of certified biocompatibility meant it couldn't be used in the final cell culture version. Tough 2000 Resin provided biocompatibility for skin contact and non-cytotoxicity with greater rigidity, but this wasn't a key factor in our final design. BioMed Clear Resin was selected as the primary material for all bioreactor components. This resin is specifically formulated for biomedical applications with proven cytotoxicity testing, can be autoclaved using standard protocols (134°C for 20 minutes or 121°C for 30 minutes), provides adequate transparency for monitoring, and possesses sufficient mechanical strength for assembly procedures including tapping and drilling. The choice of BioMed Clear Resin ensured the bioreactor met all functional requirements while maintaining compatibility with standard laboratory sterilization protocols, enhancing accessibility and reproducibility across different research settings.

SolidWorks: Computer-Aided Design Bioreactor Model

The majority of this project involved the ideation, design, and optimization of a 3D perfusion bioreactor to model the interaction between different cells in the GI tract. For this, our team utilized the SolidWorks software (2025 version).

The bioreactor for this project required computer-aided design (CAD) for 3 distinct parts: a well, a lid, and a base. Multiple wells should be able to connect via nozzles and tubing aligned on a base and fitted with lids capable of sealing the system aided by a gasket and bolts. Nozzles,

bolts, and gasket material were made readily available by our mentors. Thus, the task remained to design these three parts to make a modular and scalable design that would prove to be functional, leak-free, and optimized to reduce perfusion media and resin material waste. For this, we aimed to make a bioreactor big enough to hold the scaffold in between the two nozzles, with thick enough walls for material integrity, but avoided larger-than-necessary dimensions.

For our final model, we used three wells, three lids, and a base with the appropriate three slots. Each well had 4 holes, two on each side, to allow for bi-directional media perfusion; these holes were to be fitted with EBK-10 model 10-32 UNF nozzles (Pneumadyne) for 1/16" inner diameter tubing. Each lid contained four holes, matched at the base, to accommodate for #8 stainless steel bolts that would hold down the wells in place and apply pressure. Each of these holes then had to be tapped appropriately to create the necessary threading for nozzles and bolts. This was first performed using an industrial tapping machine for earlier prototypes, then with a hand-tapping set, referencing the Little Machine Shops's *Tap and clearance drill sizes* chart. A gasket, or o-ring, was also needed for a tight seal between the cylindrical wells and the lid. For this, we used gasket material sheets 1.59 mm thick which was cut to fit the outer diameter of the cylindrical walls; the o-ring was placed on top of the well and between the lid. The seal was made by placing the bolts through the lid and base.

Silk Solution Preparation

Our team also prepared the silk solution that was used to make the scaffolds for our finalized bioreactor product. To make this solution, the following protocol was followed:

1. Weigh out 5 grams of silk. Silk fibers were used, but cocoon pieces work as well.
2. Bring 2 liters of deionized water (DIW) to a boil. Add 4.24 grams of sodium carbonate.

3. Add silk to water and boil for 30 minutes (30mb). Occasionally pull apart fibers and stir during this process.
4. After 30 minutes have elapsed, perform 3 washes with DIW to remove any remaining sericin before letting silk dry overnight by laying flat on aluminum foil.
5. Record mass (in grams) of dried silk. Measure out four-fold in milliliters of a 9.3 M LiBr solution and add to silk in a 20 mL beaker. Leave in the oven at 60℃ for an hour, stirring at the 30 minute mark.
6. Prepare 3.5 kDa molecular weight cut-off (MWCO) dialysis tubing, clamping both the top and bottom of the tube once the silk solution has been added.
7. Add tube to 5 L DIW. Perform between 5-8 water changes; monitor solution conductivity to track frequency needed for water changes.
8. Once most of the LiBr solution has been removed, centrifuge the silk solution twice in 50 mL tubes at 9000 RPM at 4℃ for 20 minutes per cycle. Filter through 40 μm mesh between cycles to remove possible contaminants and non-dissolved silk.

Silk Sponge Scaffold Preparation

Our team was interested in using the silk solution to prepare porous silk sponge scaffolds with a lumen for media perfusion through the middle. These scaffolds were based on approximately 20 mm long polydimethylsiloxane (PDMS) molds, with a metal rod inserted through the middle to create the lumen. The silk solution's concentration was determined via a water weight evaporation test; the desired concentration was 6% silk fibroin. For solutions that are slightly dilute (under 6%), the approach would be to evaporate some of the water, whereas solutions that are too concentrated can be diluted by the addition of DIW. The solution was slowly added to the mold via micropipetting to avoid bubble formation. If bubbles are formed, it

is best to use a vacuum pump to extract any trapped air to maintain the integrity of the scaffold. The mold was then stored upright in the freezer overnight. It was then lyophilized, or freeze dried, to form a dry, porous, sponge-like scaffold structure.

The dry scaffolds were then sterilized by autoclaving (at 121 °C for 20 minutes). This serves the purpose of sterilization to eliminate any microorganisms present as well as induction of Beta-sheet formation; the high heat and moisture induce the crosslinking of the silk protein which makes the scaffold insoluble in aqueous solutions and increases its mechanical stiffness and stability. After autoclaving, we hydrated the scaffolds by placing them in DIW until they were ready for use.

Perfusion and Leakage Testing

After all of this, it was possible to conduct perfusion analysis tests. Media to grow Caco-2 cells, a human colon cell line which would have been a target for us to model with the reactor, is costly and requires a lot of time and resources to make; instead, since cell seeding fell out of the scope of our project timeline, our team replaced cell growth media with DIW. The final model was assembled with the appropriate tubing and the hydrated scaffolds were inserted to fit over the barbed nozzles before sealing the wells shut. Time constraints limited the possibility of adding cells to our model for testing, but the simulated perfusion was nonetheless performed to characterize the integrity of the bioreactor.

RESULTS

The 3D design of the bioreactor took many iterations to obtain a reasonable, feasible, and working model. The finalized, major dimensions for each part are listed below for reference:

Table 1. Bioreactor Model Major Dimensions

Cylindrical wells	Lid	Base
Outer diameter: 40.0 mm	Length & width: 25x25 mm	Length & width: 210x50 mm
Wall thickness: 5.0 mm	Circular diameter: 40.3 mm	Height: 10.0 mm
Height: 30.0 mm	Wall/hole center distance: 6.5 mm	Well spacing: 40.0 mm

Using parts designed with these dimensions, the following 3D assembly within SolidWorks was obtained:

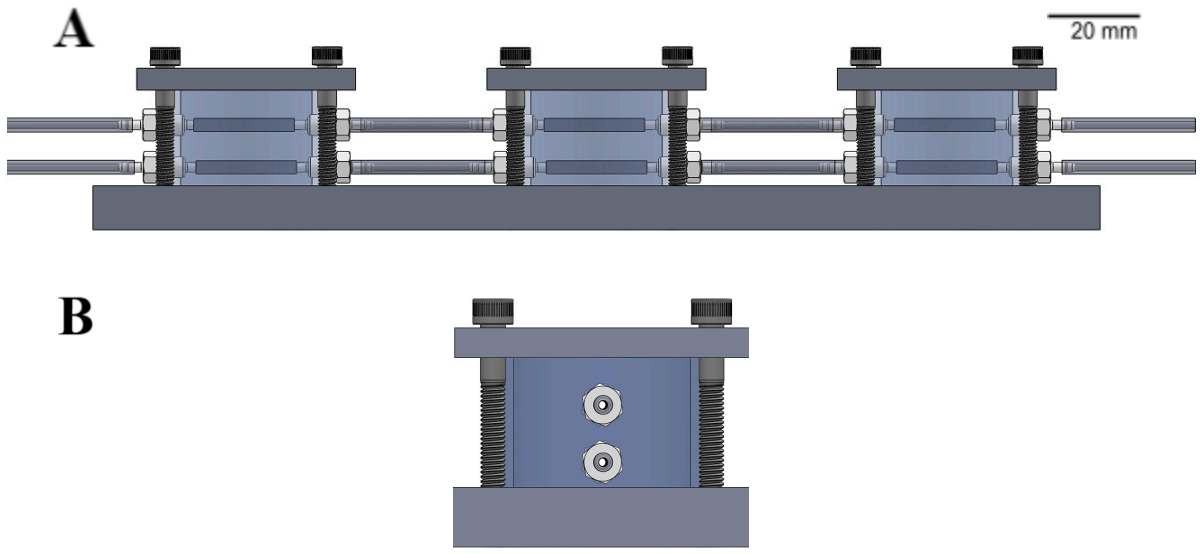


Figure 1. SolidWorks Assembly of Perfusion Bioreactor. (A) Side View. (B) Front View.

The assembly made it possible to confirm the model's feasibility and the compatibility of all the parts. The wells were designed in a cylindrical fashion in such a way that there would be enough printing tolerance with the base and lid for the latter to successfully accommodate the well diameter. The diameter of the well was retrofitted based on the known size of the scaffold (~20 mm), accounting for wall thickness and inward nozzle protrusions.

The parts were printed using Formlabs Form 3B+ stereolithography (SLA) printer using their BioMed Clear Resin. The parts were printed and confirmed to have enough tolerance to fit

well with each other. For this final version, holes were already printed into the design, so there was no need for drilling (unlike in the first working prototype). Tapping for threading of the lid bolts and the nozzles was performed using size 29 and 21 drill bits, respectively. This allowed for smooth integration of these parts into the assembled model.

Following the outer diameter dimensions for the cylindrical well, the diameter for a matching gasket was measured (40 mm) and cut to size. This was placed between the well and the lid before closure with the lid bolts, with enough tolerance built-in to the model's dimensions to allow for a snug fit.

The silk scaffolds were successfully prepared from the described 30mb silk solution procedure and by pouring it into the PDMS molds and lyophilizing, autoclaving, and hydrating the sponge scaffolds.

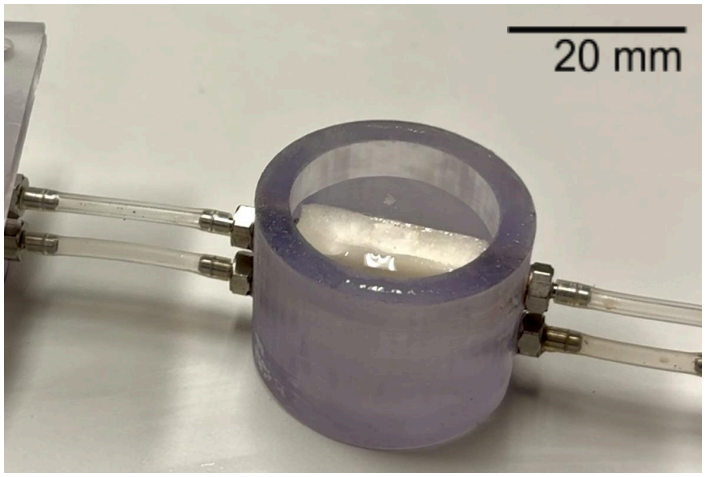


Figure 2. Sponge silk scaffolds prepared from 30 minute boil 6.4 % silk fibroin solution successfully attached to the bioreactor model during perfusion testing

The scaffolds maintained their structural integrity post-sterilization and hydration, while still remaining porous and flexible. The length of the scaffolds was confirmed to be 20 mm, which was within the accepted range for the printed design.

The bioreactor was successfully assembled to match the 3D assembly model and was used for a perfusion test using water flowing through the wells.

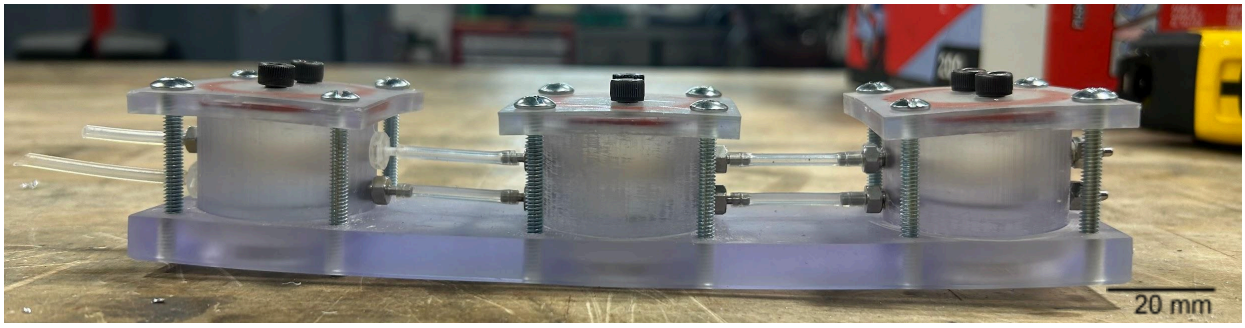


Figure 3. Finalized assembled bioreactor model containing silk sponge scaffolds capable of functional water perfusion

The assembled bioreactor system was confirmed to be leak-free and fully functional. The three cylindrical wells were successfully connected to one another via tubing and dual flow was achieved. Tapping of holes on the lid allowed for the installation of the same EBK-10 nozzles as the wells (not pictured above, but screws were used as placeholders), which in a running model would allow for gaseous exchange at each individual site.

DISCUSSION

The objective of this project of designing, printing, and assembling a biocompatible, modular bioreactor to model the complex interactions between different segments in the gastrointestinal tract was successfully met. The final device demonstrates several key features that address limitations in existing *in vitro* GIT models, including modularity for scalable experimental design, bidirectional flow capability for simulating physiological perfusion, biocompatibility for cell culture applications, optical transparency for non-invasive monitoring, and customizable operating conditions.

The iterative design process proved essential for developing a functional bioreactor system. The use of SolidWorks for computer-aided design enabled rapid prototyping and refinement of component dimensions, allowing our team to optimize critical parameters such as wall thickness, well height, and port positioning before committing to final fabrication. The modular approach, with separate components for the cylindrical wells, lids, and base plate, provided flexibility in troubleshooting and making improvements to individual elements without requiring complete system redesign.

Our use of 3D printing with the Formlabs Form 3B+ printer and BioMed Clear Resin proved advantageous over traditional machining approaches. This manufacturing method offered benefits including reduced material waste compared to traditional machining, faster time for prototype iterations, accessibility for students without extensive machine shop experience, and the ability to create complex geometries that would be challenging to machine. The biocompatibility and sterilizability of the BioMed Clear Resin addressed critical requirements for cell culture applications, while the material's transparency enabled visual confirmation of scaffold positioning and flow characteristics. This makes our prototype a powerful, viable, and scalable alternative to the existing machined version of this bioreactor system.

The assembly process required careful attention to creating leak-free connections between components. The tapping and drilling procedures for creating threaded connections proved more challenging than anticipated with 3D-printed parts compared to traditionally machined materials. The resin material, while providing adequate strength, required precise control during tapping to avoid cracking or deformation. This is why it was important to gain familiarity with tap and drill charts to understand with precision the hole sizing required to save time and trouble in the machine shop when it came to this. To troubleshoot, our team opted for

printing the holes into the model, knowing that they wouldn't come out perfect due to the printer's inevitable imperfections, but tapping over them instead. This proved to be an effective and time-saving fix for our project.

The sealing system, utilizing gasket material between the cylindrical wells and lids, combined with bolted compression, was effective in preventing leaks during perfusion testing. The four-bolt configuration provided uniform pressure distribution around the circumference of the well, ensuring consistent gasket compression. The perfusion tests with water successfully demonstrated that the assembled bioreactor could maintain a sealed system under flow conditions, validating the design of the sealing mechanism and confirming the feasibility of using this system for future cell culture experiments with growth media.

The preparation of silk solution through degumming, dissolution in lithium bromide, dialysis, and lyophilization provided biocompatible scaffolds that would be suitable for cell attachment and tissue engineering applications. The scaffold fabrication process, while time-intensive, yielded materials with appropriate porosity and mechanical properties for supporting three-dimensional cell cultures and to withstand perfusion testing. The compatibility between the scaffold dimensions and the internal geometry of the bioreactor wells was also successfully achieved, with the barbed nozzle ends effectively securing the scaffolds in place during perfusion. Future iterations could explore different silk concentrations or incorporate other natural polymers to better mimic the specific mechanical properties of different intestinal segments in each compartment.

The primary limitation of this work was the time constraint that prevented cell seeding and long-term culture studies. While the perfusion testing with water demonstrated the mechanical functionality and leak-free operation of the bioreactor, validation with actual cell

cultures remains necessary to fully assess the system's capability for supporting intestinal tissue models. Future work should prioritize seeding Caco-2 cells or other intestinal cell lines onto the silk scaffolds, allowing sufficient time for cell attachment, proliferation, and differentiation before initiating perfusion experiments with cell culture media to confirm this bioreactor system's functionality once and for all.

The modular nature of the system presents opportunities for creating more complex multi-segmental models. Future iterations of this project could incorporate four or more connected chambers, each with segment-specific conditions (pH, oxygen concentration, flow rates) to better simulate the progressive changes that occur along the length of the GIT. Integration of sensors for pH and dissolved oxygen within each chamber would enable precise control and measurements of these parameters.

Ultimately, this bioreactor system represents an accessible and reproducible platform for studying multi-segmental GIT interactions. The use of 3D printing technology, as opposed to traditional machining, makes this type of device more readily available to research laboratories without extensive fabrication facilities. The modular design principles demonstrated here could be adapted for other organ system models requiring controlled perfusion and multi-compartment architecture. What we have designed here could be used in numerous applications including drug absorption studies across different intestinal segments, research of nutrient transport mechanisms along the GIT, modeling of inflammatory bowel diseases and other GIT conditions, and assessment of host-microbiome interactions in a controlled environment. The ability to independently control conditions in each chamber while maintaining physiological connectivity between segments makes this system particularly valuable for studying the interdependent nature of digestive processes.

CONCLUSION

In conclusion, this project successfully demonstrated the feasibility of creating a 3D-printed, modular bioreactor for multi-segmental GIT modeling. The device achieved leak-free operation, incorporated biocompatible silk scaffolds, enabled bidirectional perfusion, and provided visual monitoring capabilities. While further validation through cell culture studies is needed, the foundation established by this work provides a promising platform technology for advancing *in vitro* GIT research in a more physiologically relevant and reproducible manner.

REFERENCES

Chen, Y., Rudolph, S. E., Longo, B. N., Pace, F., Roh, T. T., Condruti, R., Gee, M., Watnick, P. I., & Kaplan, D. L. (2022). *Bioengineered 3D Tissue Model of Intestine Epithelium with Oxygen Gradients to Sustain Human Gut Microbiome.* Advanced healthcare materials, 11(16), e2200447. https://doi.org/10.1002/adhm.202200447

Chen, Y. (2025). BME175 Class Slides for Weeks 7; Tufts University, 2024.

Chen, Y., Lin, Y., Davis, K. et al. *Robust bioengineered 3D functional human intestinal epithelium*. Sci Rep 5, 13708 (2015). https://doi.org/10.1038/srep13708

Costa, J., & Ahluwalia, A. (2019). *Advances and current challenges in intestinal in vitro model engineering: A digest*. Frontiers in Bioengineering and Biotechnology, 7, 144. https://doi.org/10.3389/fbioe.2019.00144

García Mendez, D. F., Sanabria, J., Wist, J., & Holmes, E. (2023). *Effect of Operational Parameters on the Cultivation of the Gut Microbiome in Continuous Bioreactors Inoculated with Feces: A Systematic Review*. Journal of agricultural and food chemistry, 71(16), 6213–6225. https://doi.org/10.1021/acs.jafc.2c08146

Little Machine Shop. (n.d.). *Tap and clearance drill sizes.* [Web page] https://littlemachineshop.com/reference/tapdrill.php

O'Brien, F. J. (2011). *Biomaterials & Scaffolds for Tissue Engineering*. Materials Today, 14 (3), 88-95. https://doi.org/10.1016/S1369-7021(11)70058-X

Singh R. P. (2024). *Progress and challenges in designing dynamic in vitro gastric models to study food digestion*. Frontiers in nutrition, 11, 1399534. https://doi.org/10.3389/fnut.2024.1399534

Sumeri, I., Arike, L., Adamberg, K., & Paalme, T. (2008). *Single bioreactor gastrointestinal tract simulator for study of survival of probiotic bacteria*. Applied Microbiology and Biotechnology, 80(2), 317-324. https://doi.org/10.1007/s00253-008-1553-8

Thermo Fisher Scientific. (n.d.). *Cell Culture Basics Handbook*. Gibco Education. https://assets.thermofisher.com/TFS-Assets/BID/Handbooks/gibco-cell-culture-basics-handbook-guide.pdf

Ward, J. B. J., Keely, S. J., & Keely, S. J. (2014). *Oxygen in the regulation of intestinal epithelial transport*. The Journal of Physiology, 592(12), 2473-2489. https://doi.org/10.1113/jphysiol.2013.270249